

LAN-EX H/F Tester — Operator Manual

This guide is for the operator running the throughput test. It covers how to start the tester, enter the setup, read the live screen, and find your results. No technical background is needed.

1. What this tool does

The LAN-EX H/F Tester checks that each **Head/Field (H/F) pair** in a set can move data fast enough and hold a steady connection. It runs continuously — testing every pair over and over — until you stop it. When you stop, it writes a report showing whether each pair **PASSED** or **FAILED**, stamped with your initials and the date/time so the result is accountable.

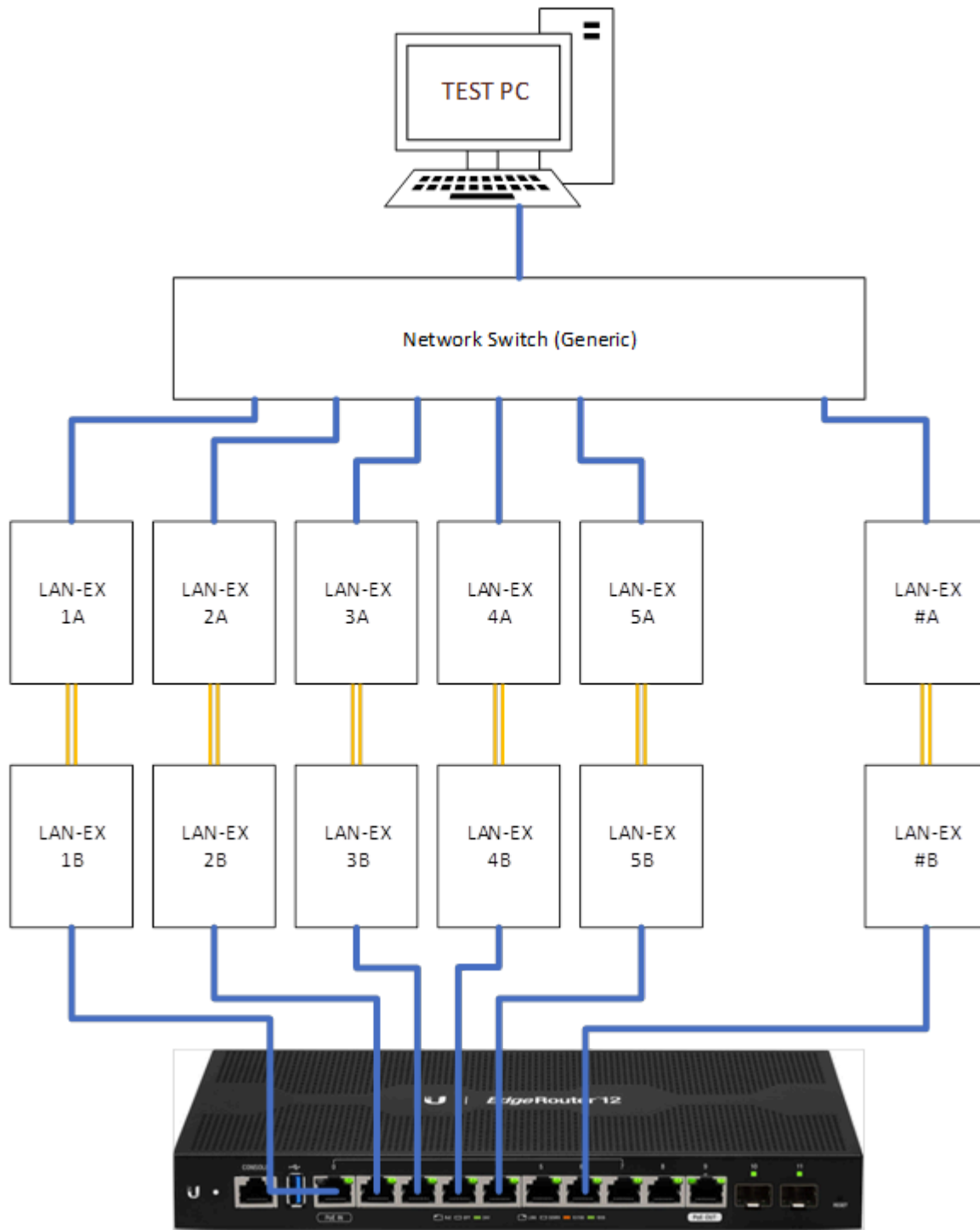
The test runs in two phases, repeated each cycle:

- **Phase A — Max throughput.** Each pair is tested one at a time, first sending (TX) then receiving (RX). Each direction is measured against a target speed.
 - **Phase B — Soak.** All pairs run together at a low, capped speed for a while. This phase only watches for **dropped connections**.
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2. Before you start

Have these ready:

- **Your initials** (2–4 letters) — recorded on the result.
- **The pair serial numbers**, one per pair, in the form **1234567 - 7654321** (7 digits, a dash, 7 digits).
- All pairs **cabled and powered**.



3. Starting the tester

From the tester PC, open a terminal in the tester folder and run:

```
make run
```

(or `./LAN-EX-Tester` if it is already built).

The setup wizard starts on **Step 1 / 4**.

Tip: At almost any step you can type **b** and press **Enter** to go **back** and fix the previous answer. The tester checks every entry and will politely re-ask if something looks wrong — you can't accidentally continue with a bad value.

4. Setup steps

You'll be walked through these, one screen at a time. Type the answer, then press **Enter**.

Step	Screen	What to enter	Rules
1/4	Operator initials	Your initials	2–4 letters
2/4	Number of pairs	How many H/F pairs	1 to 8
3/4	Pair serial numbers	Each pair's serial	NNNNNNN-NNNNNNN — 7 digits, dash, 7 digits
4/4	Dip switch check	Confirm the dip settings with y	See the on-screen diagram

Entering serial numbers: you enter them one at a time. After the last one, the tester lists them all and asks you to **confirm**:

- Press **y** to accept the list.
- Press **n** to re-enter the **last** one. (To fix an earlier one, use **b** to step back.)

The pair-serial screen shows a header like **Pairs in this set: 3** — glance at it to make sure the pair count is what you expect **before** the test starts.

5. Pre-test connectivity check

After setup, the tester **pings every pair** to make sure it can reach them.

- If all pairs answer, the test begins automatically.
- If a pair does **not** answer, you'll see: *"Pair X could not pass the preliminary test — Please check its connections."* Fix the cabling/power for that pair, then **press any key to try again**.

6. Running the test — the live screen

Once connectivity passes, the live monitor appears and the test runs continuously:

```
LAN-EX H/F Tester                                LIVE  press Q
to stop & report

Operator GS   Config 3 pairs   Cycle 4   Elapsed 00:14:32
-----
Phase A  max throughput - one pair at a time, 10s each direction
-----
```

```

-----
Pair Serial          TX Mbps    RX Mbps    Drops    Status
1    1234567-7654321    941        939        0    done
2    2233445-5544332    938        942        0    done
3    3344556-6655443    610         -        0    running
4    4455667-7766554    drop        -        2    FAIL
5    5566778-8877665     -         -        0    waiting

Pair 3 TX [#####-----] 62%  6s / 10s
-----
-----
Totals  done 2  fail 1      target >= 90 / 190 Mbps

```

Reading the screen:

- **Top bar** — **LIVE** means the test is running. The reminder shows **press Q to stop**.
- **Info line** — your initials, the configuration, the current **cycle** number, and how long the run has been going (**Elapsed**).
- **Phase line** — whether you're in **Phase A** (throughput) or **Phase B** (soak).
- **Pair table** — one row per pair, updating live:
 - **TX Mbps / RX Mbps** — the measured speed. Green = met the target, red = below target.
 - **Drops** — how many **cycles** this pair failed to hold a working connection in — whether it dropped mid-test or couldn't connect at all — over the whole run. Stays 0 (dim) for a healthy pair; turns **red** the moment it fails. Any drop = FAIL.
 - **Status** — what the pair is doing right now (see colors below).
- **Progress bar** — the countdown for the current step (Phase A direction, or the Phase B soak).
- **Totals** — how many pairs have finished (**done**) or failed (**fail**) this cycle, plus the target speeds.

Status colors and words:

Shows	Color	Meaning
waiting	dim	not started yet
running	teal	being tested now
retry 2/3	amber	a connection couldn't start; the tester is retrying automatically
done	green	finished this cycle successfully
FAIL	red	this pair failed (missed its target speed, or its connection dropped)

The **retry N/M** amber marker means a measurement didn't start and the tester is trying again on its own — no action needed. If all retries fail, the pair is marked **FAIL** and the run keeps going.

Stopping the test: press **q** (or **Q**) at any time. The test finishes what it's doing, stops cleanly, and moves on to the results.

Let the test run through **as many cycles as you need** for confidence. The longer it runs, the more chances each pair has to reveal an intermittent problem.

7. Understanding the result

Results are reported **per pair** — there is no single all-or-nothing verdict.

A pair PASSES only if, in every cycle it ran, it:

1. met **both** the TX and RX target speeds in Phase A, **and**
2. **never dropped** its connection (in Phase A or Phase B).

A pair FAILS if, in any cycle, it:

- came in **below target** on TX or RX, **or**
- **dropped its connection even once**. A single drop fails the pair.

When the test stops you'll see the **Run Summary**, for example:

```
Started: 2026-07-09 14:22:07
Ended:   2026-07-09 15:07:19
Cycles completed: 118

Pair 1 (1234567-7654321): PASS    peak TX 941 / RX 942 Mbps    drops: 0
Pair 2 (2233445-5544332): FAIL    peak TX 30 / RX 80 Mbps    drops: 0
Pair 3 (3344556-6655443): FAIL    peak TX 900 / RX 910 Mbps    drops: 3

1 / 3 pairs passed
```

drops: is the number of cycles the pair had no working connection (it dropped, or couldn't connect at all). Any value above 0 fails the pair (Pair 2 above failed instead on speed, with no drops). The **Connection drops** and **Errors** sections below break down which kind.

Press any key to save the reports and exit.

8. Where the reports go

Reports are saved automatically when you stop. They live in the **reports/** folder:

- **Summary report** — **reports/<N>pairs_<pair-serial>_<operator>_<date_time>.txt** (e.g. **reports/3pairs_1234567-7654321_GS_20260713_140000.txt** — how many pairs, the unit tested, who ran it, and when) Contains your initials, the configuration, the **start** → **end time**, cycles completed, the per-pair PASS/FAIL breakdown, the "X / Y pairs passed" rollup, and a list of any **connection drops** and **errors** that occurred — so every failure is documented.
- **Engineering report** — **reports/eng/eng_<...>.txt** The detailed raw measurement logs, for troubleshooting.
- **reports/allReports.txt** — a running log with every run appended.

Each run gets its **own timestamped file**, so a new run never overwrites an old one.

9. Troubleshooting

You see...	What it means	What to do
"Invalid initials / number / serial"	The entry didn't match the required format	Re-enter as prompted (the message says the expected format)
"Pair X could not pass the preliminary test"	That pair didn't answer the ping	Check its cable and power, then press a key to retry
A pair stuck on amber retry N/M	Its connection couldn't start; the tester is retrying	Wait — it resolves itself, or fails after the last retry
A pair goes red FAIL with drop	The connection dropped during the test	Note the pair; it's recorded in the report. Check cabling/hardware for that pair
A pair's TX or RX shown in red	The speed was below target	The pair fails on throughput; recorded in the report
The screen won't stop	You may not have pressed q on the test screen	Press q (or Q) while the live monitor is showing

10. Quick reference — keys

Key	Where	Does
Enter	Setup	Submit your answer
b	Setup	Go back one step
y / n	Confirmations	Accept / re-do
q or Q	Live test	Stop the test and save reports
any key	Result / error screens	Continue

Target speeds, phase durations, the soak cap, and the retry count are set by your site administrator in `config/targetBandwidth.conf`. If the targets look wrong for your equipment, raise it with them rather than changing it yourself.